Quiz: 24:-

(C25) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^2}{\sqrt{n^5 - n + 5}}$

↓ Alternating! we can use A.S.T

Alternating Series test:-

Let $a_n = \frac{n^2}{\sqrt{n^5 - n + 5}}$

Conditions:-

① is a_n positive on $[1, \infty)$? yes, because

$\frac{n^2}{\sqrt{n^2 - n 5}} = \frac{\text{Positive}}{\text{Positive}} = \text{Positive}$, where n is positive

② is a_n decreasing on $[1, \infty)$?

yes, $a_{n+1} = \frac{(n+1)^2}{\sqrt{(n+1)^5 - (n+1) + 5}} < \frac{n^2}{\sqrt{n^5 - n + 5}}$

③ is $\lim_{n \to \infty} a_n = 0$?

$\lim_{n \to \infty} \frac{n^2}{\sqrt{n^5 - n + 5}} \Rightarrow 0$ by applying limit

because of $\sqrt{n^5} > n^2$.

$\therefore \sum_{n=1}^{\infty} (-1)^{n+1} \dfrac{n^2}{\sqrt{n^5 - n + 5}}$ is Converges Absolutely by A.S.T

C26

$$\sum_{n=1}^{\infty} \frac{n^4 2^n}{(n+2)!}$$

Using Ratio test :-

$$\Rightarrow \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{(n+1)^4 \, 2^{n+1}}{(n+3)!} \cdot \frac{(n+2)!}{n^4 \, 2^n} \right|$$

$$\Rightarrow \lim_{n\to\infty} \left| \frac{(n+1)^4}{n^4} \frac{2}{(n+3)} \frac{2^n}{2^n} \frac{(n+2)!}{(n+3)!} \right|$$

$$= \lim_{n\to\infty} \left| \frac{(n+1)^4 \, 2}{n^4 (n+3)} \right| = \lim_{n\to\infty} \frac{2(n+1)^4}{n^5 + 3n^4} = 0$$

$L = 0 < 1$ because $n^5 > n^4$.

$\therefore \sum_{n=1}^{\infty} \dfrac{n^4 2^n}{(n+2)!}$ is converges by Ratio test.

(27. $\sum_{n=1}^{\infty} \frac{3^n (x-2)^n}{n}$

Using Ratio test :-

$$\Rightarrow \lim_{n\to\infty} \left| \frac{3^{n+1} (x-2)^{n+1}}{n+1} \cdot \frac{n}{3^n (x-2)^n} \right|$$

$$= \lim_{n\to\infty} \left| \frac{3(x-2)}{n+1} \cdot \frac{n \cancel{3^n} \cancel{(x-2)^n}}{\cancel{3^n} \cancel{(x-2)^n}} \right|$$

$$= \lim_{n\to\infty} \left| 3(x-2) \right| \frac{n}{n+1} = |3(x-2)| < 1$$

$$\Rightarrow -1 < 3(x-2) < 1$$

$$\Rightarrow \frac{-1}{3} < x-2 < \frac{1}{3}$$

$$\Rightarrow \frac{-1}{3} + 2 < x < \frac{1}{3} + 2 \Rightarrow \frac{5}{3} < x < \frac{7}{3}$$

Check end points :-

when $x = 5/3$

$$\lim_{n\to\infty} \frac{3^n (x-2)^n}{n} = \lim_{n\to\infty} \frac{3^n (5/3 - 2)^n}{n}$$

$$= \lim_{n\to\infty} \frac{3^n (-1/3)^n}{n} = \lim_{n\to\infty} (-1)^n \frac{1}{n}$$

$= \lim_{n\to\infty} (-1)^n \frac{1}{n}$ Alternating Series $a_n = 1/n$

① a_n is positive where n is positive and positive on $[1,\infty)$

② a_n is decreasing because $a_{n+1} < a_n$

$$\frac{1}{n+1} < \frac{1}{n}$$

③ $\lim_{n\to\infty} a_n = 0$? yes $\lim_{n\to\infty} \frac{1}{n} = 0$

$\therefore \lim_{n\to\infty} (-1)^n \frac{1}{n}$ is Converges by A.S.T

When $x = 7/3$

$$\sum_{n=1}^{\infty} \frac{3^n (x-2)^n}{n} \Rightarrow \lim_{n\to\infty} \frac{3^n (7/3 - 2)^n}{n}$$

$$= \lim_{n\to\infty} \frac{3^n (1/3)^n}{n} = \lim_{n\to\infty} \frac{1}{n} = 0$$

So:-

Interval of convergence is $[5/3, 7/3)$

Radius of convergence is $R = 1/3$